

Deciphering Archaeological Spatial Structure on Domestic Sites

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Archaeological Spatial Structure

Two Components of *Archaeological Spatial Structure**

- ***Site Structure***: the spatial distribution of subsurface features across sites (*e.g.* remains of structures, fences, pits, etc).
- ***Intra-Site Artifact Distributions***: the spatial distribution of artifacts an across sites ~ artifact density surfaces.
- Wandsnider, LuAnn, 1996, Describing and Comparing Archaeological Spatial Structure. *Journal of Archaeological Method and Theory* 3(4): 319-384.
<https://digitalcommons.unl.edu/anthropologyfacpub/70/>

A Little History

- Serious interest in spatial structure is relatively recent.
- *Culture History (c. 1920s to 1960s)*. Primary focus: The distribution of “cultures” in time and space. Sites are "homogeneous artifact mines."
- *New or Processual Archaeologists (post 1960's)*. How cultures function is important too, *e.g.* social and social organization. Archaeological spatial structure offers essential clues

Archaeological Spatial Structure

If we want the evidence from archaeological spatial structure to be useful, we have to make inferences about the causal dynamics responsible for the spatial patterns .

We need to think about two sets of processes.

Systemic context processes*

- *Site structure*: what factors affect the spatial organization of activities and the facilities or non-portable artifacts (whose archaeological traces are “features”) with which activities are associated?
- *Artifact Distributions*: what factors affect where artifacts are used and discarded?

Archaeological context processes

- Site Structure: What factors affect the ways in which facilities are represented as the features we see in the record? (e.g. preservation)
- Artifact Distributions: What factors affect spatial patterns after artifact deposition and site abandonment? (e.g. plowing, erosion,)

* Schiffer, M.B., 1972: Archaeological context and systemic context, *American Antiquity* 37, 156–65. See also... Furlan G. 2024: Revising the concepts of systemic context and archaeological context: a proposal. *Archaeological Dialogues*. 31(2):208-219.

Systemic Context Processes

Spatial organization of activities and the facilities with which they are associated: a simple model.

Imagine a domestic site where all activities (e.g eating, sleeping, food preparation, bulk processing, storage) all happen in the same place.

The extent to which one activity will be spatial segregated from others is a function of

- Benefits of moving scale inversely with interference potential:
 - activities take up lots of space
 - activities are dangerous
 - activities require uninterrupted time

Higher interference potential means activities are more likely to be moved farther away.

- Costs of moving the interfering activity
 - Greater distance means greater travel costs to and from
 - Greater access frequency also increases travel costs.

General vs. Special Activity Areas

General Activity Areas

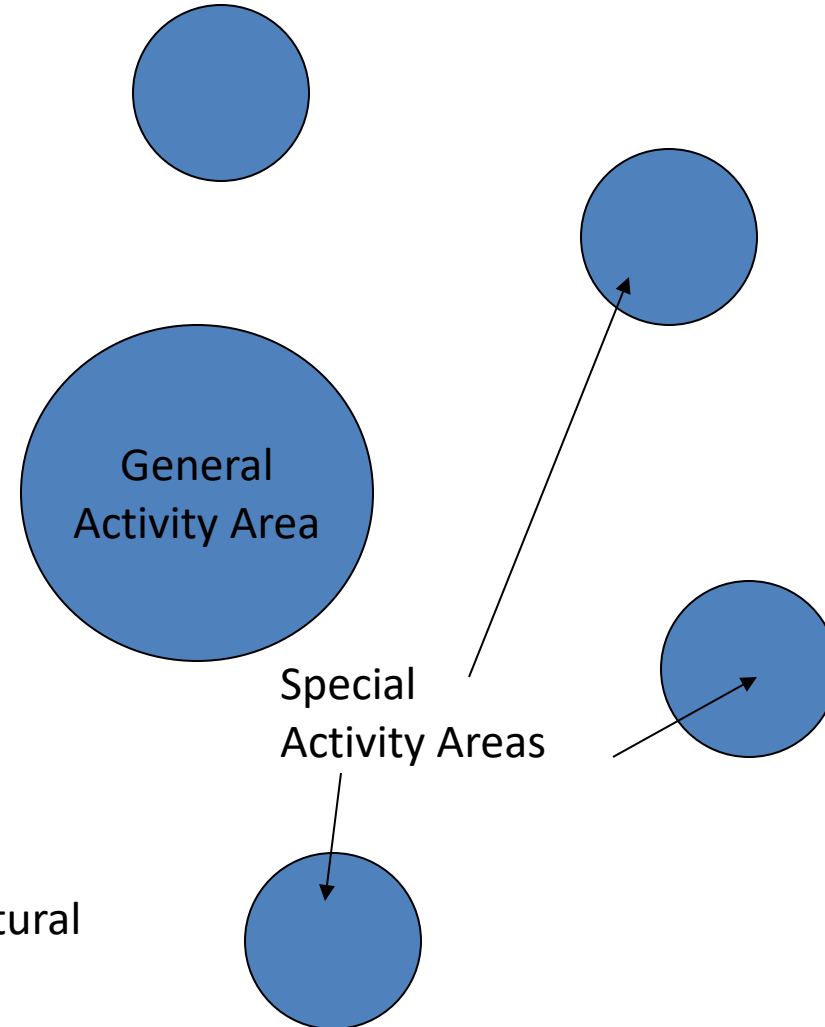
- sleeping, eating, socializing, cooking
- sleeping, cooking often separate within them

- Family (kin based)
- Communal (cross cut kin lines e.g. gender based)

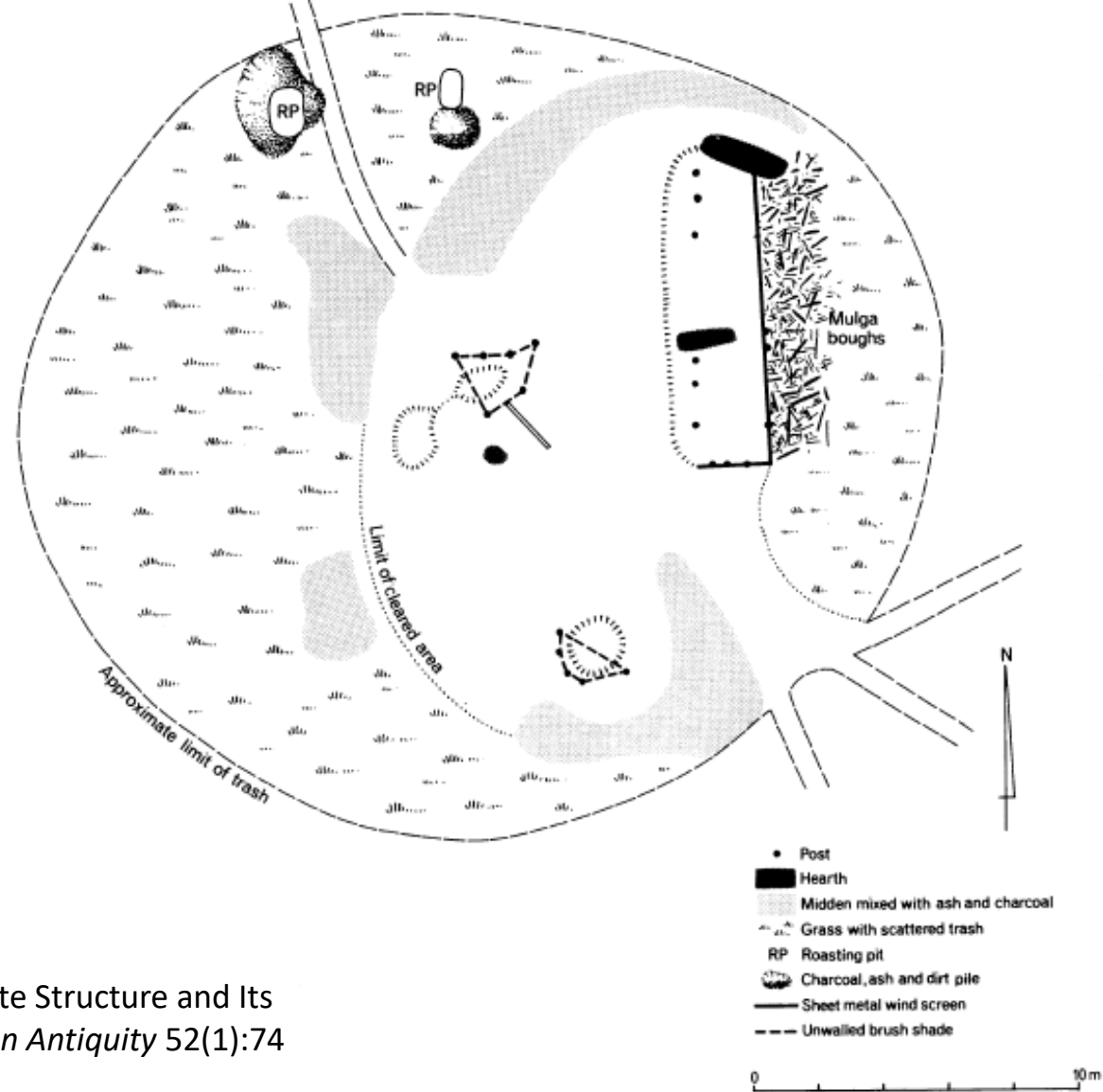
a.k.a, “nuclear” (John Yellen)
“intensive” (Lewis Binford)
“household” (O’Connell)

Special Activity Areas

- ecologically variable
- e.g. kangaroo roasting, car repair
- cooking for large numbers (signals)
- bulk processing and storage of agricultural produce
- lots of refuse, space, time



Implications for Site Structure



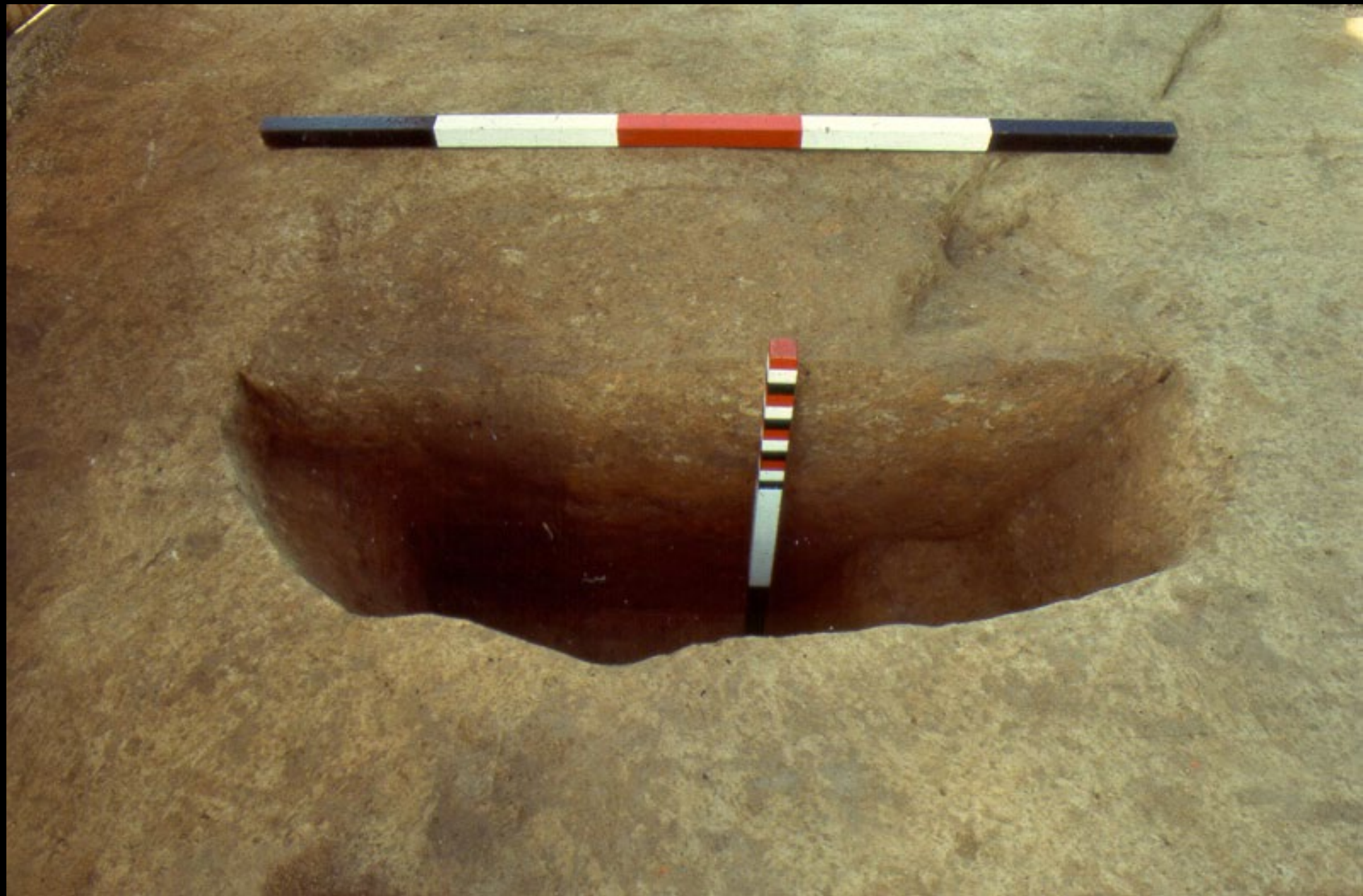
Source:

O'Connell, James, 1987, "Alyawara Site Structure and Its Archaeological Implications. *American Antiquity* 52(1):74

Figure 1. Plan of Alyawara household activity area, adjacent roasting pits, and refuse disposal zone.

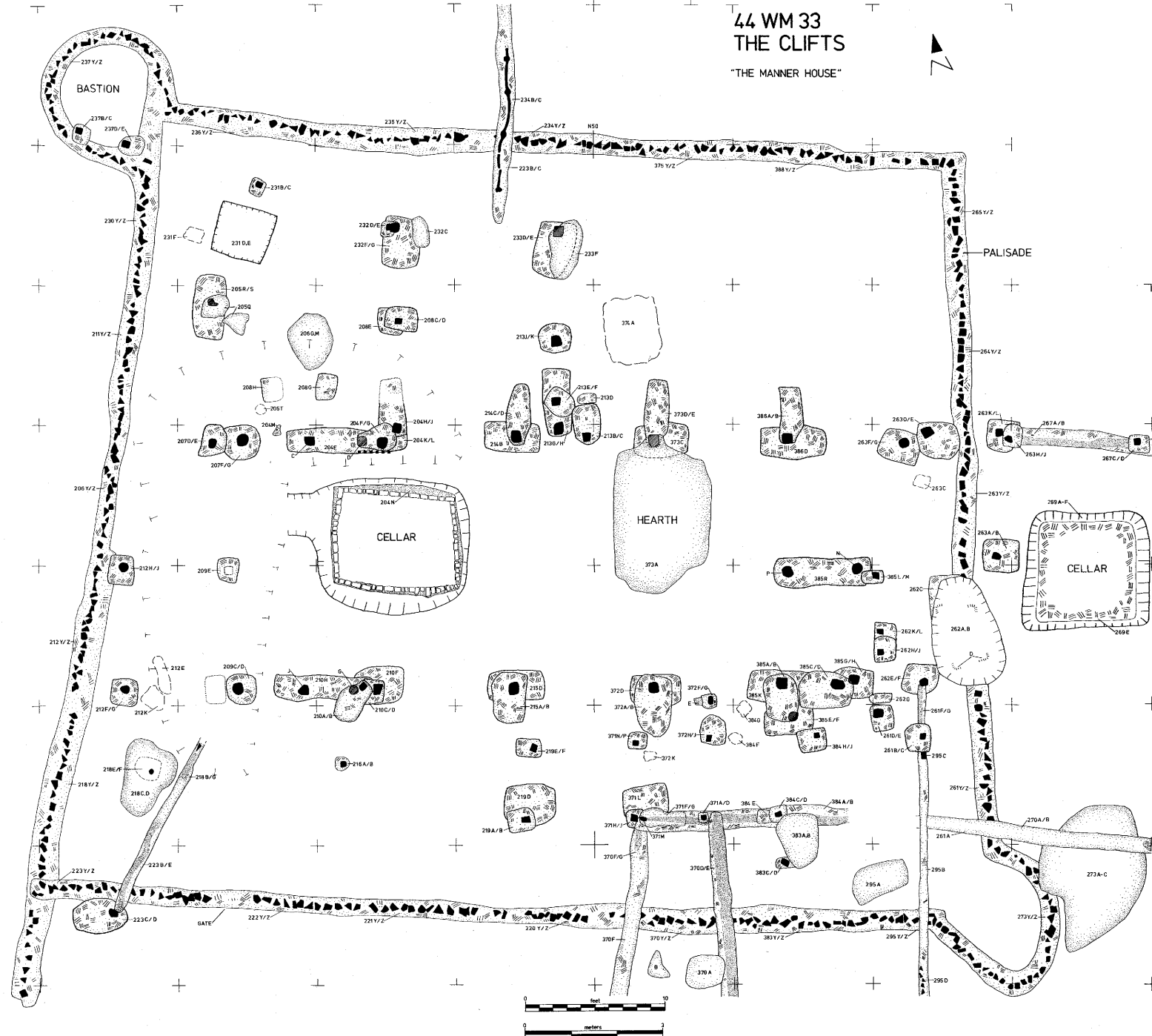


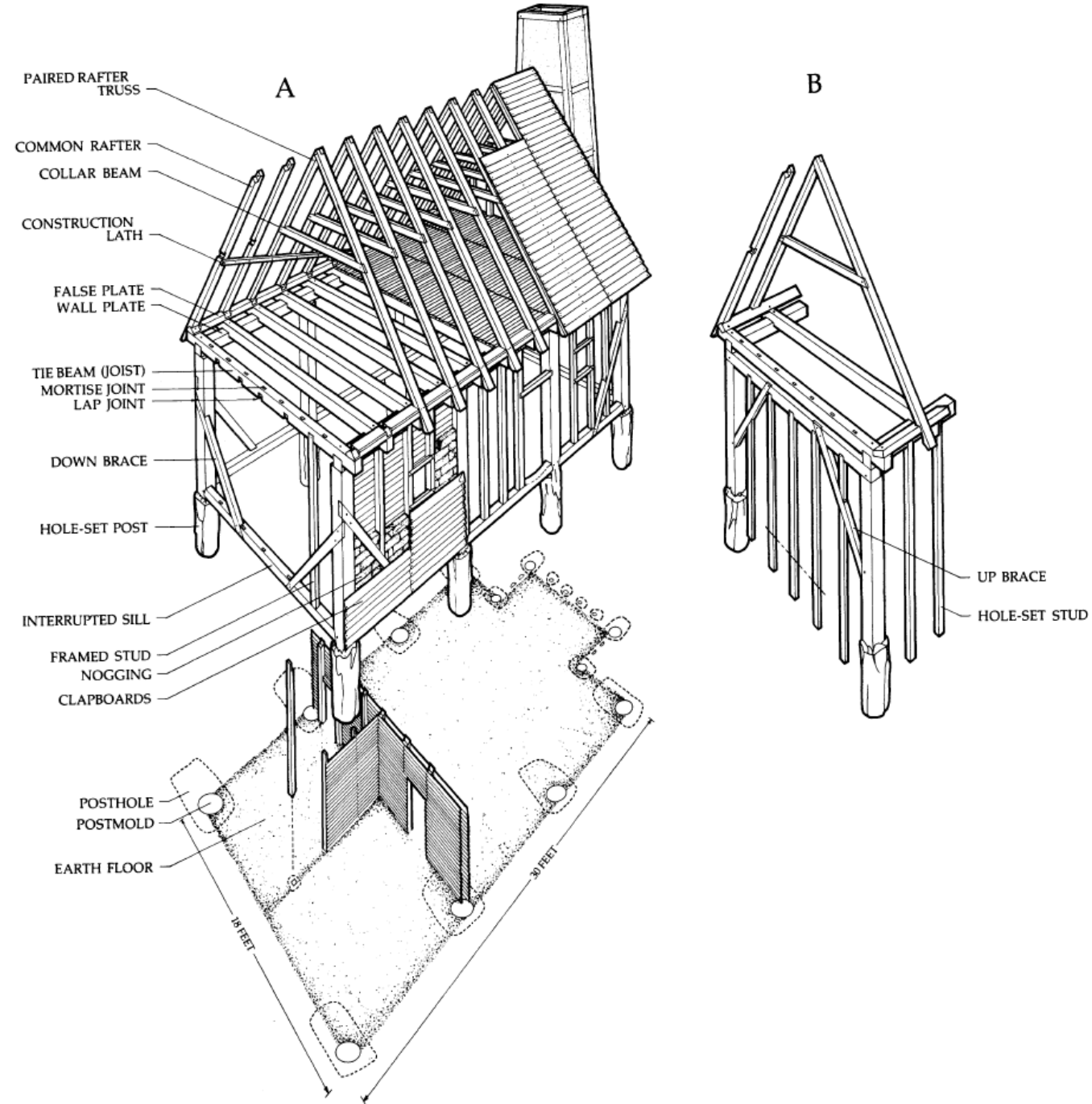




44 WM 33
THE CLIFFS

"THE MANNER HOUSE"









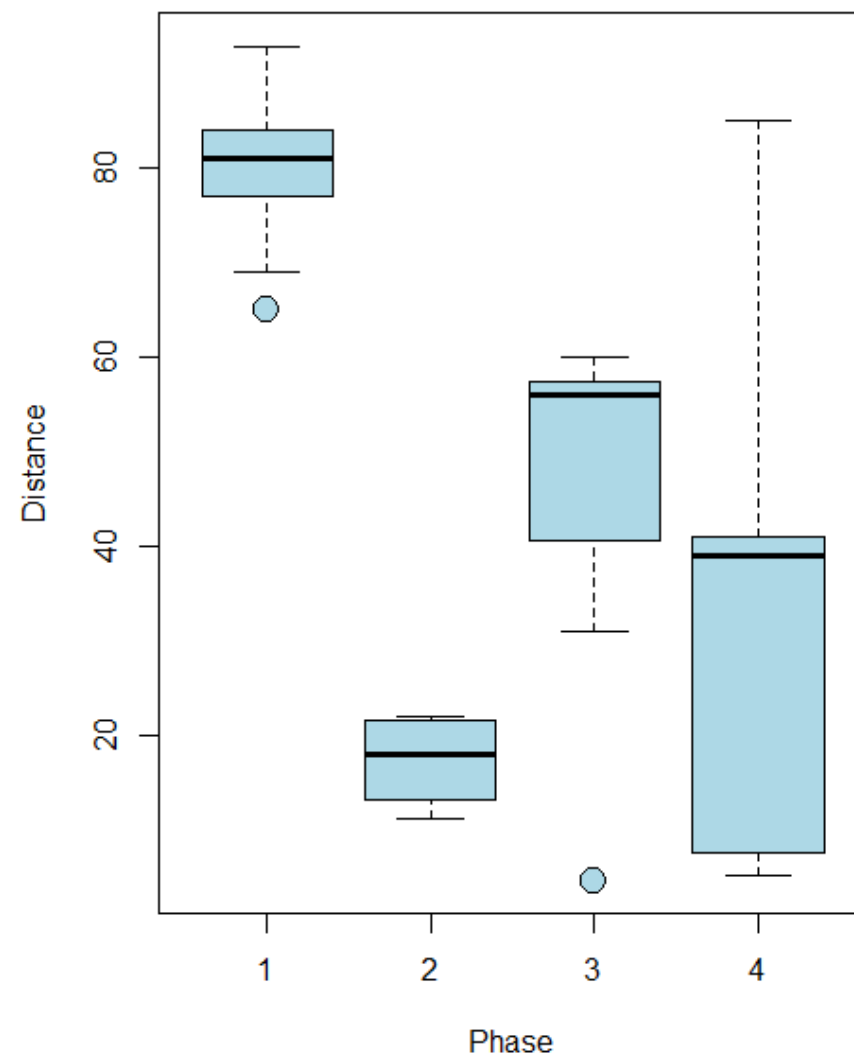
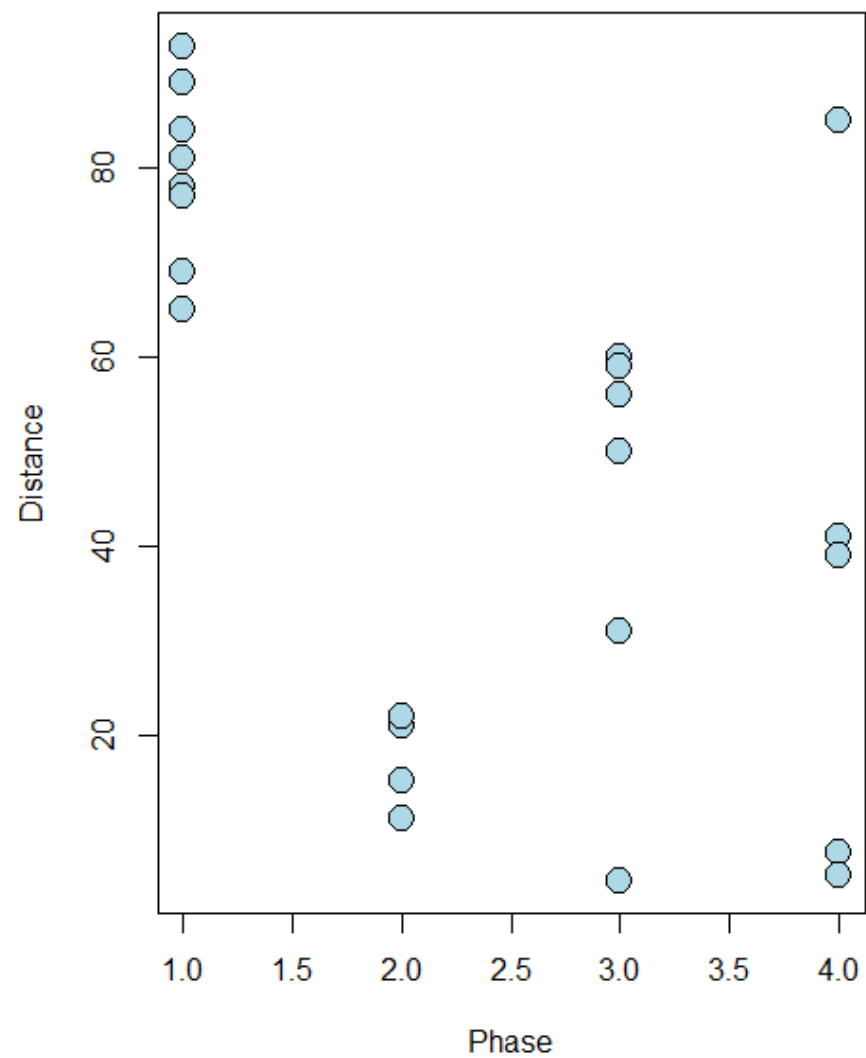




Context	Distance	Length	Width	Depth	Phase	
305A_G	78		10.7	9.7	1.8	1
304A_C	93		7.3	3.5	1.1	1
346A_D_362A_C	84		8.2	5	0.9	1
351B_H	69		10.2	3	1.3	1
356A_C	77		3.7	1.5	0.3	1
390M_N	65		6.2	4.2	0.8	1
360A	81		2	2	0.3	1
290A_D	89		3.5	3	0.6	1
289A_H	84		10.5	5.5	1.7	1
250D_E	21		7.5	6.4	0.7	2
273A_C	15		10	8	1.2	2
274A_B_E_G	22		11	7	1.2	2
295A	11		4.1	2.3	1	2
205G_M	4.5		4	3.1	0.6	3
240F_G	31		4.5	3.1	0.6	3
259A_D	56		5.5	4.5	0.7	3
288C_R	60		12	8.5	1.3	3
288S_AD	59		13.5	4.4	1.2	3
255A_E_Z_AH	50		13.5	10.5	1.1	3
255F_Y	56		5.6	4.5	2	3
309A	56		5.5	5	0.6	3
226B	7.5		4	3	0.3	4
277A_C	41		11.5	4.7	0.5	4
280A_H	39		10	7.4	1.1	4
383A_B	5		3.9	3	0.5	4
345A_D	85		5.4	4.4	1.1	4

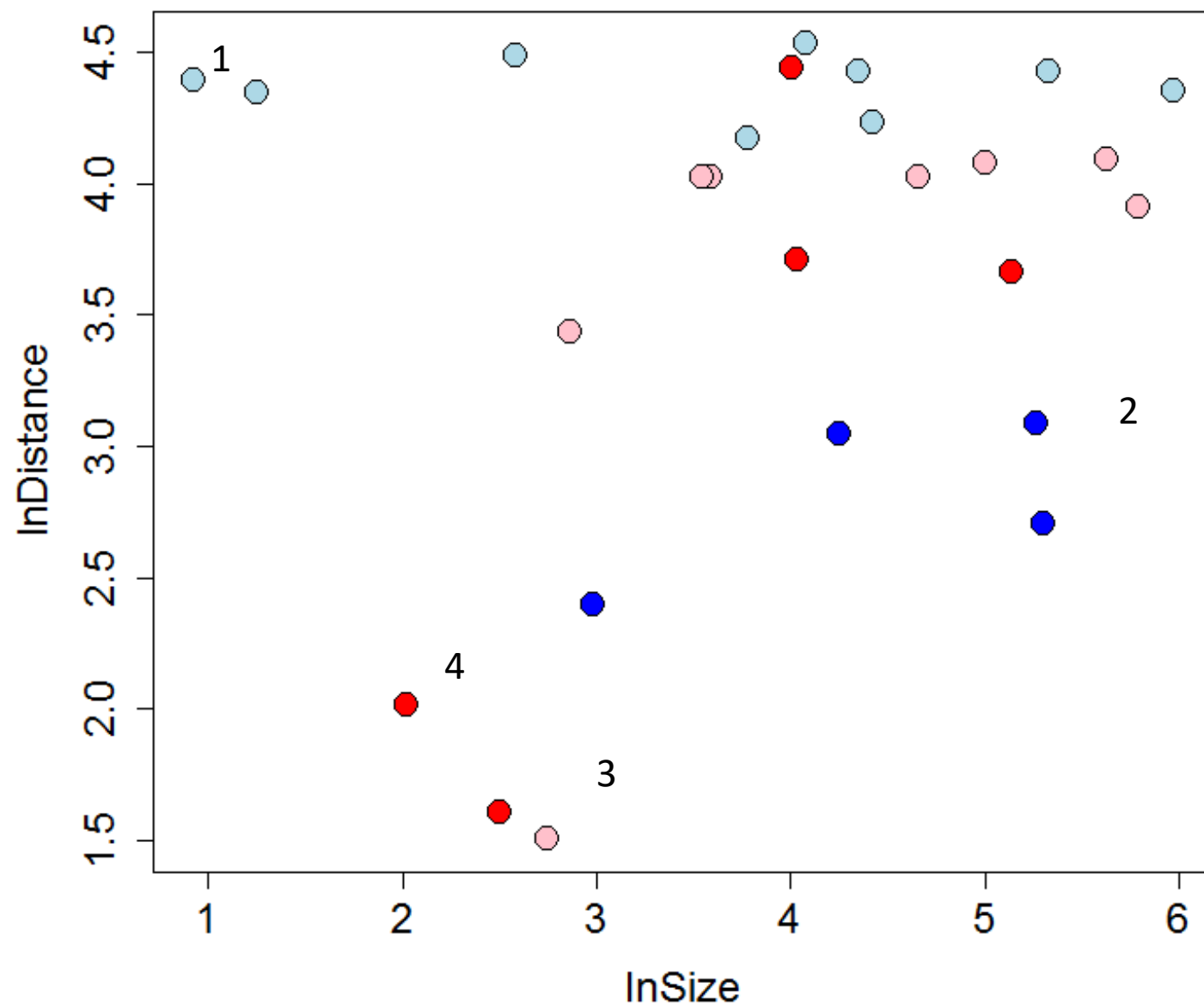
Clifts Plantation

Distances of pits from the main house



Clifts Plantation

Pit Distances vs. Pit Size (log scale)

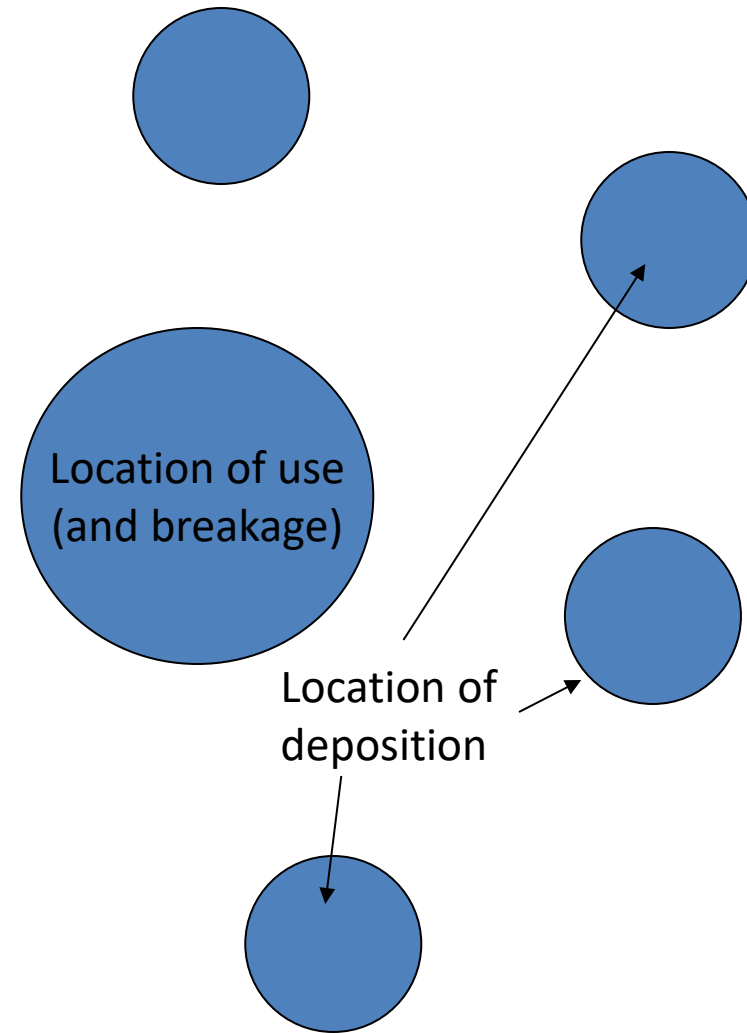




Systemic context processes

Spatial organization discard

- ***The Pompeii Premise***: the *false* idea that the location of artifact discard is the location of use and breakage.
- ***Primary refuse*** vs. ***secondary refuse*** (Mike Schiffer)
- *Cleaning up* creates some patterns and destroys others
- How far broken/failed artifacts are transported effects spatial scale at which spatial patterns can be interpreted.
- ***Secondary refuse nearly always has useful spatial pattern at some informative scale!!***



Systemic context processes

Spatial Organization of Discard.

The distance that artifacts will be transported from the location of use and breakage is a function of...

- Benefits of cleaning up
 - Scale inversely with interference potential with other activities
 - Amount of artifacts ~ occupation span, number of occupants
 - Artifact size: large artifacts are more likely to interfere.
 - Hazard potential. *e.g.* glass, poop
 - Importance of “other activities”
 - *Higher interference potential means refuse is more likely to be moved farther away.*
- Costs of cleaning up and moving the refuse
 - Scale with distance moved
 - Artifact size: little bits are harder to pick up or sweep than big bits.
 - *Easy cleanup means refuse is more likely to be moved farther away.*

Systemic Context Processes

Spatial organization discard

- How to recognize cleaning up?
- Ethnoarchaeological evidence: Alyawara men's camp (O'Connell 1987)

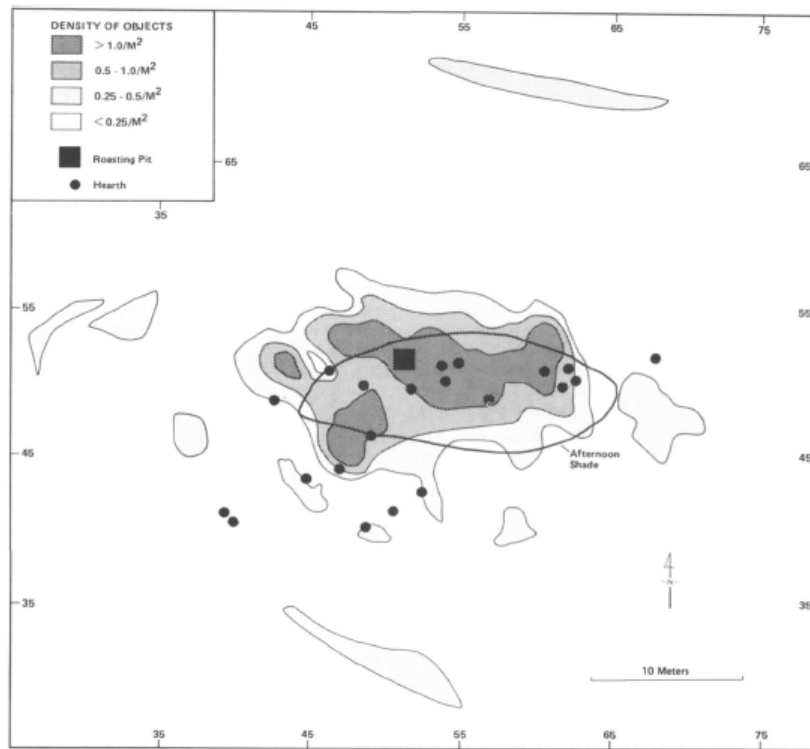


Figure 11. Contour map of refuse density at Bendaajerum *apulla* men's household activity area. Contours computed on refuse distribution data summarized by 2 m square. Total number of refuse items = 972. See Table 7 for description.

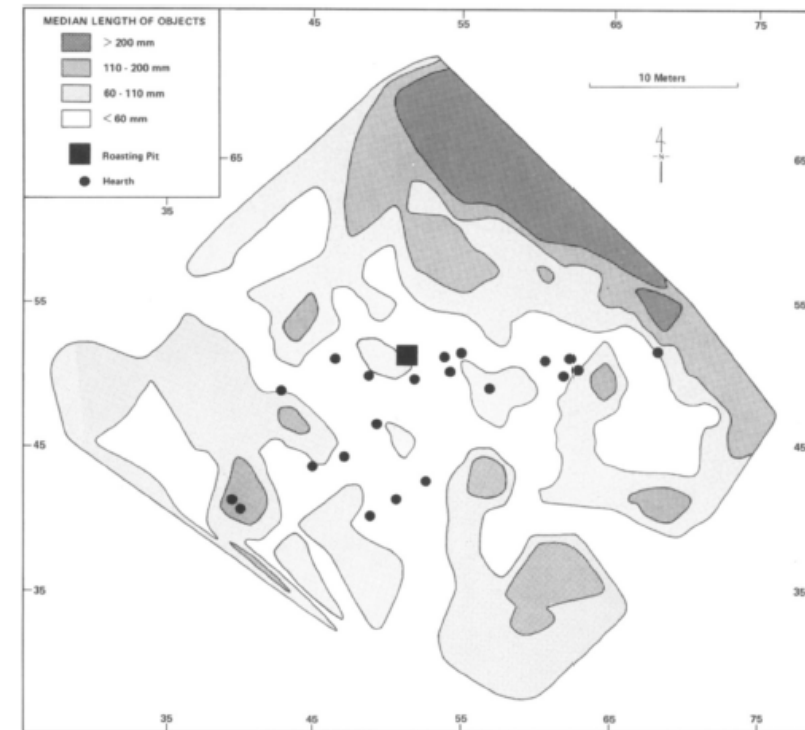


Figure 12. Contour map of median length of non-cloth objects at Bendaajerum *apulla* men's household activity area. Contours computed on data summarized by 2 m square.

Site 8 Example

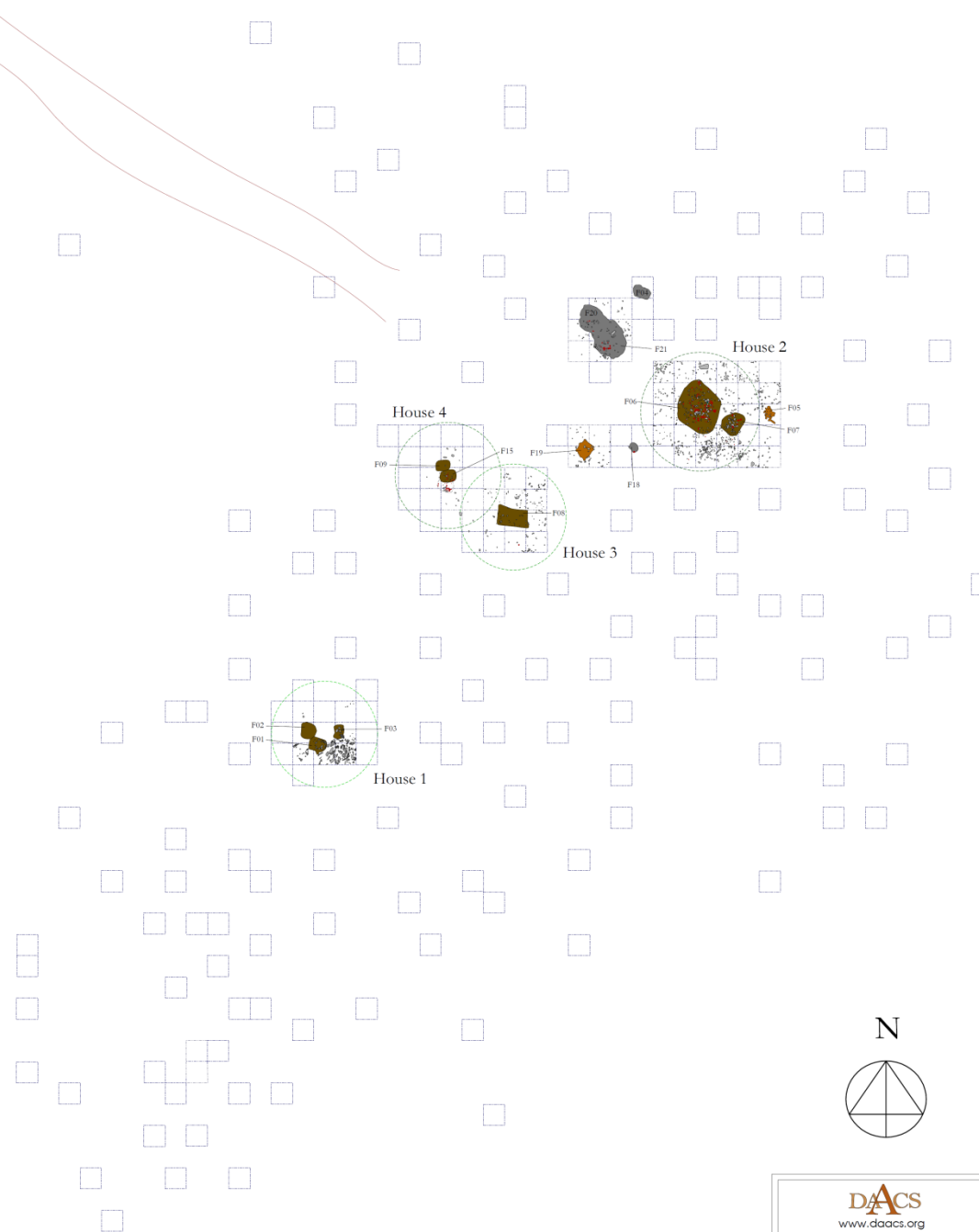
- c. 1770-1800
- heavily plowed
- stratified random sample
- 266 5-foot quadrats

0 25 50

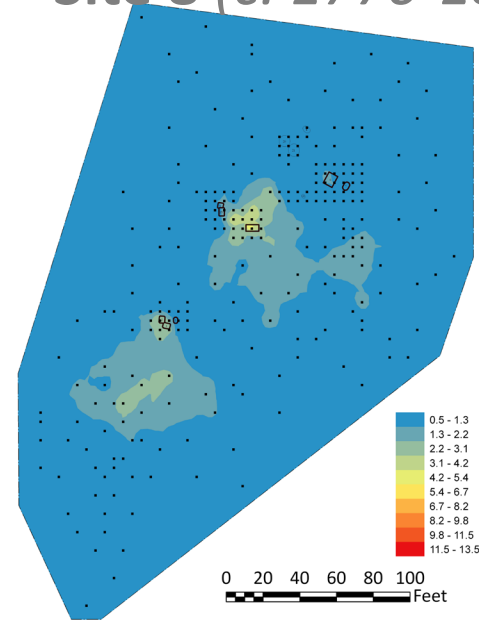


Feet

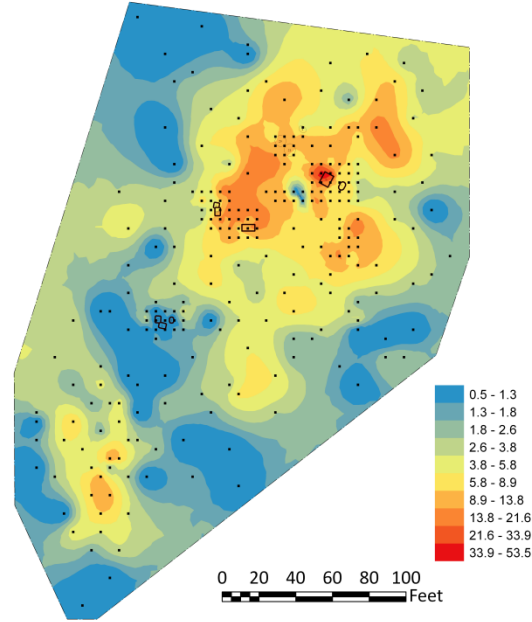
- = Suspected House Location
- = Subfloor Pit/Cellar
- = Brick
- = Stone
- = Slate
- = Misc. Feature
- = Tree Hole
- = Modern Road



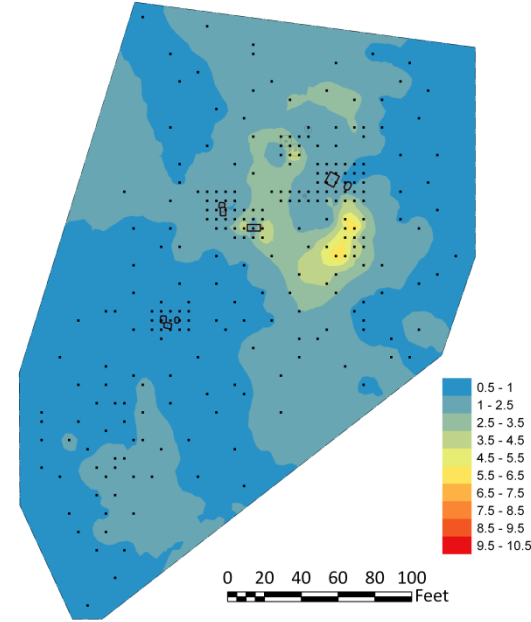
Site 8 (c. 1770-1800)



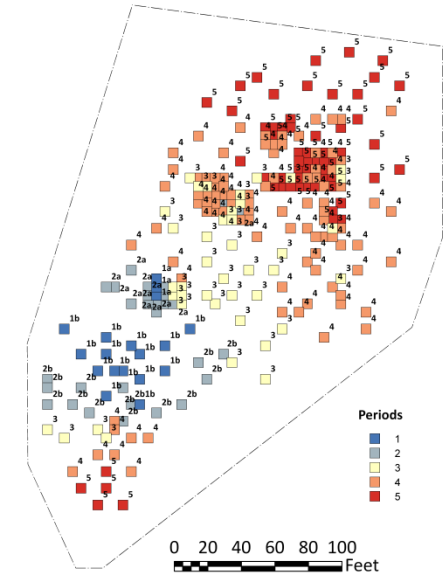
Delft (1600-1802)



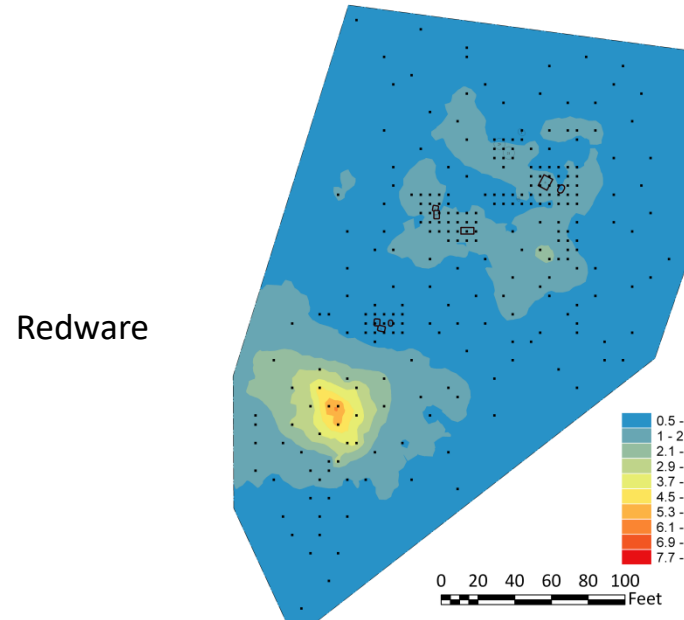
Creamware (1762-1820)



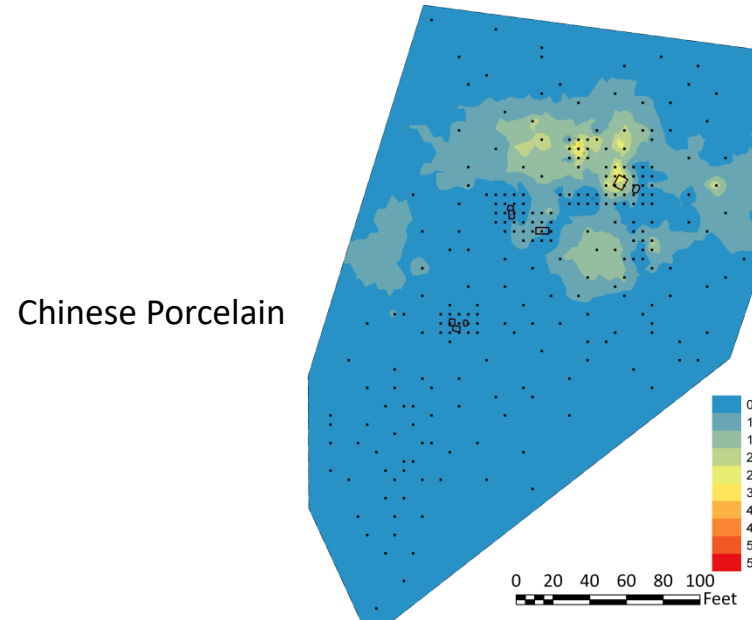
Pearlware (1775-1830)



Correspondence-Analysis Chronology



Redware

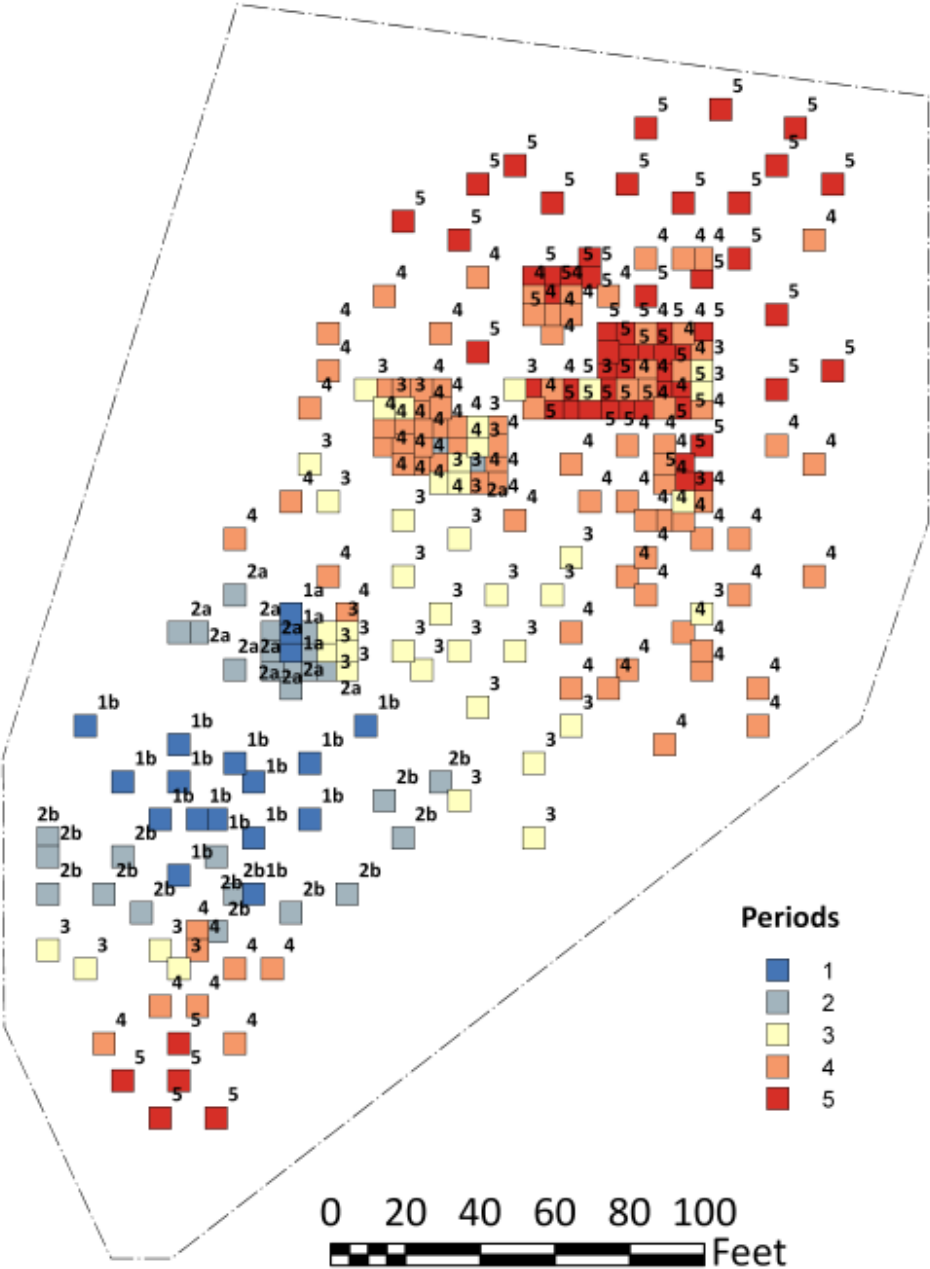


Chinese Porcelain

Artifact density surfaces interpolated from counts in scattered quads, using Kriging.

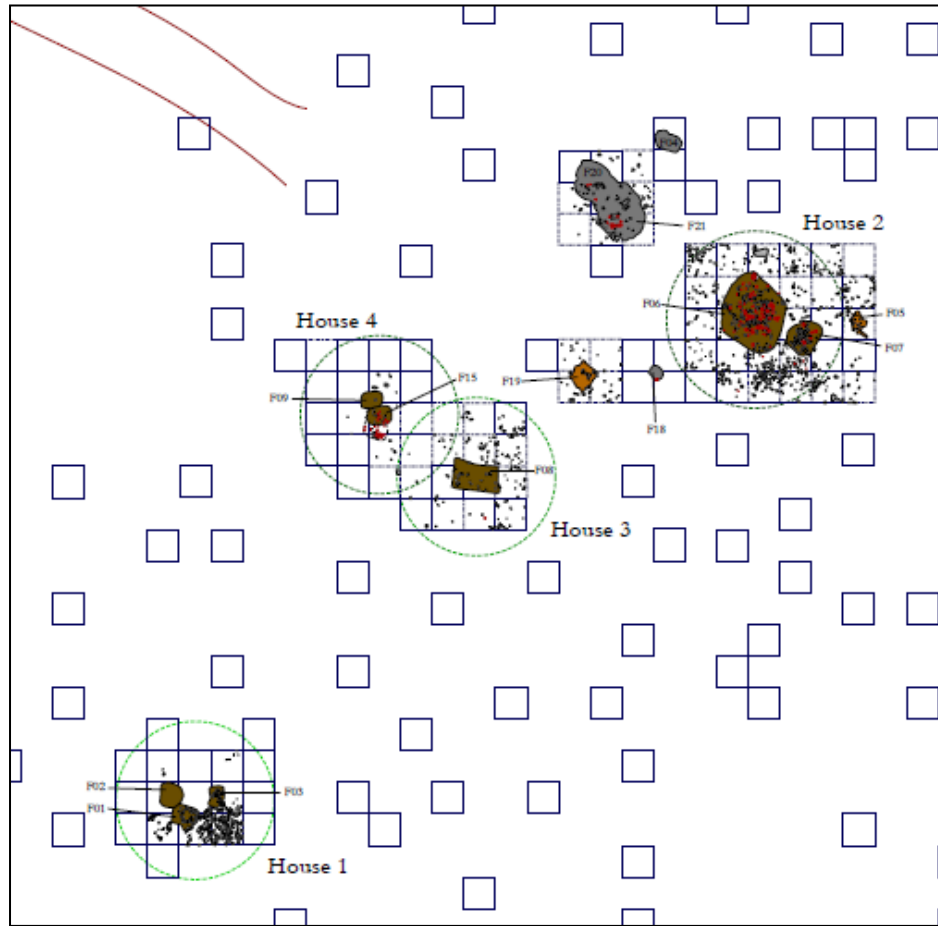
Site 8

Correspondence Analysis Chronology



Site 8 (c. 1770-1800)

Subfloor pit evidence



- Pits get larger over time: House 1 pits < House 3 and 4 pits < House 2 pits.
- House 2 has a brick-lined pit



Partially excavated sub-floor pits under Structures 3 and 4



Partially-excavated , brick-lined sub-floor pit under Structure 2

Systemic Context Processes

Spatial organization of discard

- How to recognize cleaning up?
- Ethnoarchaeological evidence: Alyawara men's camp (O'Connell 1987)

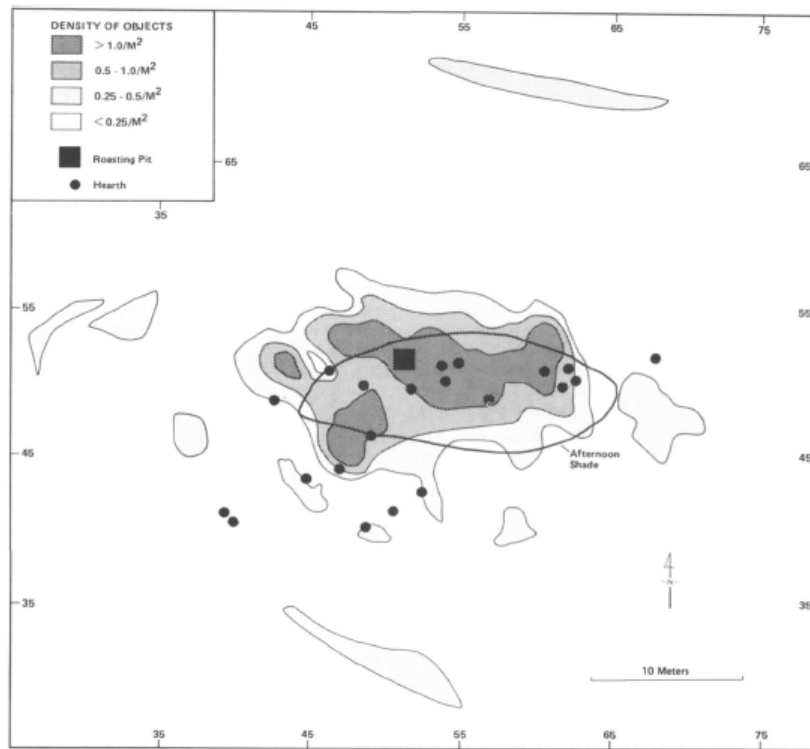


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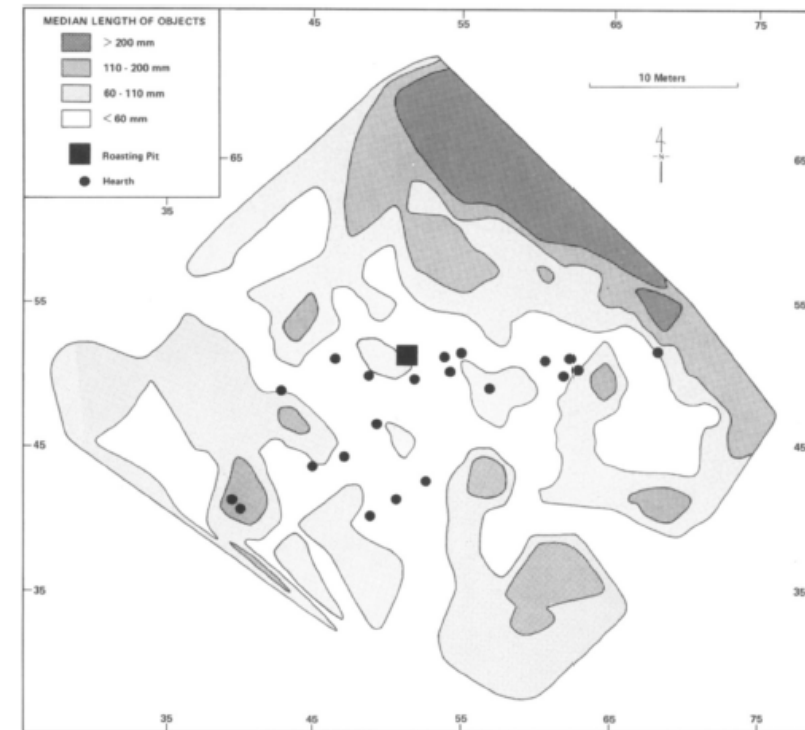


Figure 12. Contour map of median length of non-cloth objects at Bendaajerum *apulla* men's household activity area. Contours computed on data summarized by 2 m square.

The Artifact Size Index

A statistic to recognize the size sorting that results from cleaning up

$$ASI_i = \frac{(x_i - pN_i)}{\sqrt{p - (1 - p)/N_i}}$$

p = Site-wide proportion of large artifacts

x_i = Number of **large artifacts** from the i 'th quadrat

N_i = Total artifacts from the i 'th quadrat

Positive ASI: More large artifacts than expected

Negative ASI: Fewer Large artifacts than expected

So...

pN_i = **Expected number of large artifacts** in the i 'th quadrat, if all quads had the site-wide proportion.

$(x_i - pN_i)$ = **How many more (or less) large artifacts there are than expected.**

$\sqrt{p - (1 - p)/N_i}$ = Standard deviation of the *Gaussian approximation* to the *binomial distribution*, which is a simple statistical model of the chance of getting 0, 1, 2, 3 ... larger artifacts in a sample of size N_i when the underlying probability of getting a large artifact is p .

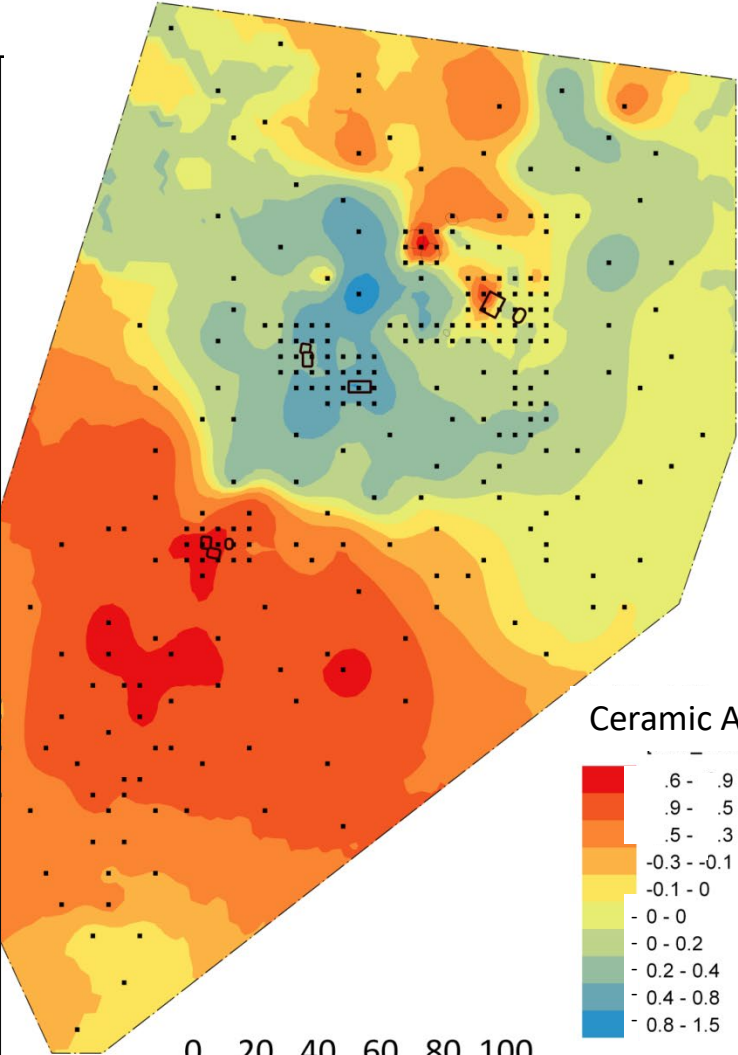
More secret sauce:

- Interpolate continuous surfaces from ASI values.

Site 8

Size Sorting: Ceramics

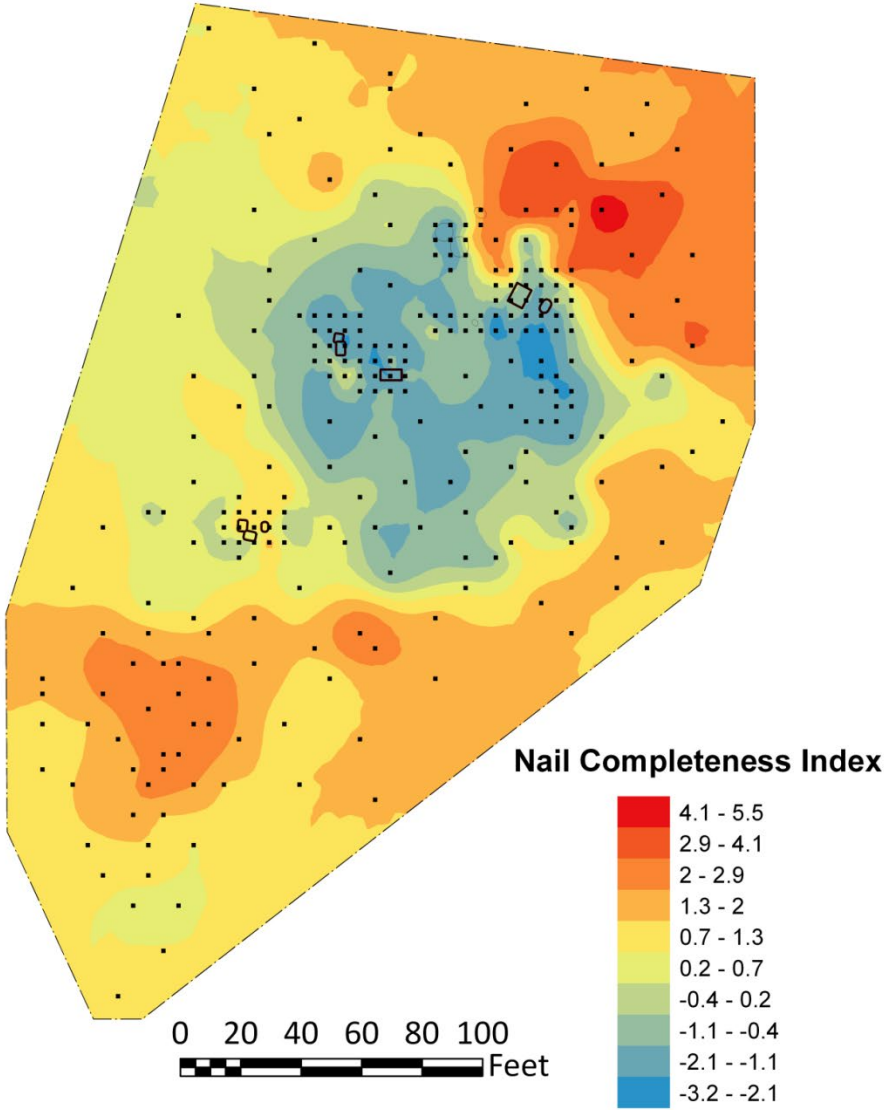
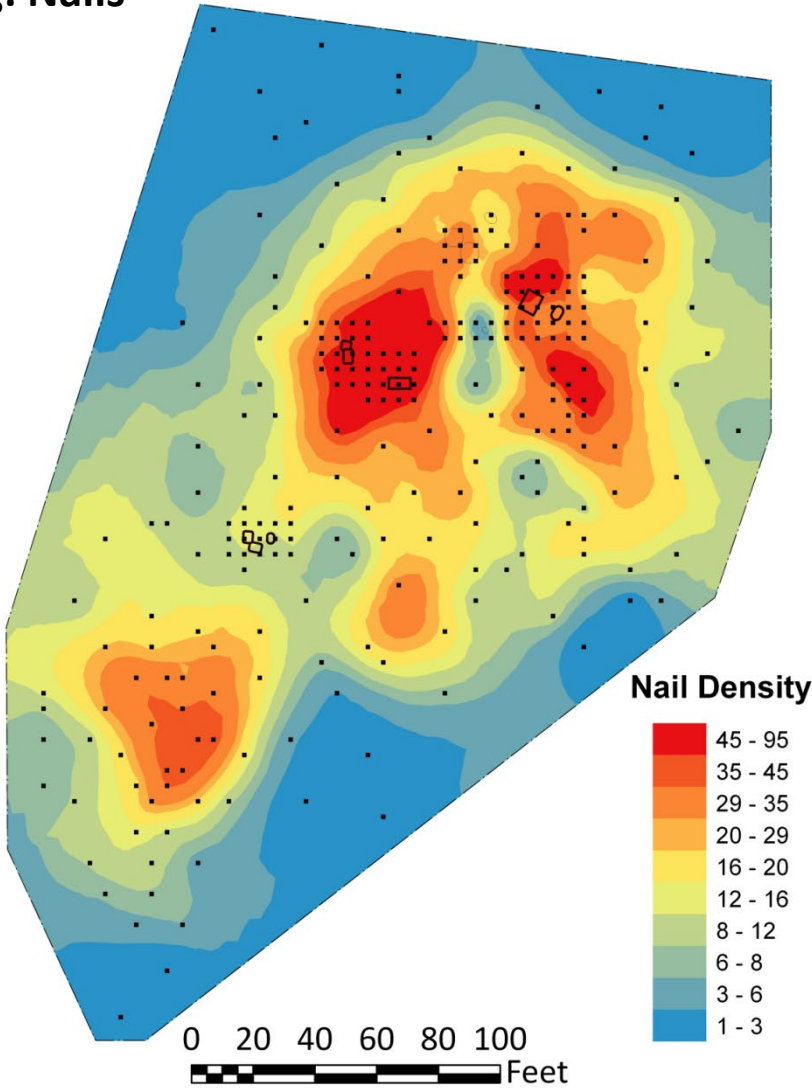
Ceramic Density



0 20 40 60 80 100 Feet

Site 8

Size Sorting: Nails



Archaeological Spatial Structure

- **Site Structure:** how subsurface features are arranged across a site — for instance, the remnants of buildings, fences, pits, and comparable traces.
- **Intra-Site Artifact Distributions:** the way artifacts are spread within a site, represented through artifact density surfaces.
- **Model of General versus Special Activity Areas**
 - Interference costs offer an incentive to separate special activity areas from general ones.
 - Cost of repeated traveling between general and special activity areas constrains that movement.
 - The framework applies to location of features and location of trash disposal (cleaning up)
- **Case Study 1: Site Structure**
 - How site structure evolved over time: the positioning of borrow pits at the Clifts Plantation Site
 - Distance of pits from the main house scales with how intensively yard around it is used: number of people, number of outbuilding.
- **Case Study 2: Intra-site Artifact Distributions**
 - Variation in level of cleaning up among households: size sorting
 - Establish a site chronology using spatial patterns
 - Variation in levels of size sorting among households scale with how intensively yards are used.